



Assignment Verification Report

Project: TelcoBot — AI-Powered Customer Support Chatbot for TelcoMax

Verification Date: 2026-05-29

1. README.md Section Checklist

#	Required Section	Status	Details
1	Project Title	✓ Present and complete	# 📡 TelcoBot – AI-Powered Customer Support Chatbot for TelcoMax (line 1)
2	Selected Use Case	✓ Present and complete	## 📌 Use Case (line 7) — Telecom customer support covering billing, plans, technical, usage, and account management
3	Tools Used	✓ Present and complete	## 🛠️ Tools & Technologies Used (line 21) — Table listing OpenAI GPT-4, LangChain, LangGraph, ChromaDB, SQLite, Gradio, LangSmith
4	APIs Integrated	✓ Present and complete	## 🗡️ APIs Integrated (line 35) — OpenAI Chat Completions, OpenAI Embeddings, LangSmith API (3 APIs total)
5	LangGraph Workflow Explanation	✓ Present and complete	## 🔗 LangGraph Workflow Explanation (line 43) — Includes Mermaid diagram and detailed step-by-step breakdown of classifier → router → agents → tools → response flow
6	Memory Implementation	✓ Present and complete	## 🧠 Memory Implementation (line 97) — Explains MemorySaver checkpoint, thread IDs, context continuity, anaphora resolution, and session management

#	Required Section	Status	Details
7	How to Run the Application	✅ Present and complete	## 🚀 How to Run (line 152) — Covers prerequisites, installation steps (clone, venv, pip install, .env setup), running the app, and optional LangSmith setup
8	Example Prompts	✅ Present and complete	## 💡 Example Prompts (line 197) — Table with 15 example prompts across 6 categories (Billing, Plans, Technical, Usage, Account, General)
9	Challenges Faced	✅ Present and complete	## ⚠️ Challenges Faced (line 263) — 5 challenges documented: tool routing loop, memory scope, RAG chunk quality, intent ambiguity, tool argument extraction
10	Future Improvements	✅ Present and complete	## 🌐 Future Improvements (line 277) — 8 improvements listed: multi-intent, auth sessions, voice, proactive alerts, HITL, deployment, evaluation, multi-language

README Verdict: ✅ All 10 required sections are present and substantive.

2. Screenshots Directory

#	File	Status	Description
1	screenshots/ 01_initial_ui.png	✓ Present	79,859 bytes — Gra- dio chat interface on launch
2	screenshots/ 02_bill_inquiry.png	✓ Present	100,278 bytes — Billing agent fetching customer bill details
3	screenshots/ 03_plan_information. png	✓ Present	112,680 bytes — Plans agent listing available plans
4	screenshots/ 04_technical_support _rag.png	✓ Present	115,686 bytes — RAG-powered troubleshooting
5	screenshots/ 05_memory_multiturn. png	✓ Present	98,645 bytes — Multi- turn memory demon- stration
6	screenshots/ 05b_conversation_con text.png	✓ Present	112,014 bytes — Context continuation across turns
7	screenshots/ 06_usage_tracking.pn g	✓ Present	98,172 bytes — Us- age agent showing consumption data

Screenshots Verdict: ✓ 7 screenshots present, all referenced in README.md.

3. Demo Video

Item	Status	Details
demo_video.mp4	✓ Present	649,828 bytes (634 KB) — Referenced in README under ## 🎬 Demo Video section

Demo Video Verdict: ✓ File exists and is referenced in README.

4. Tools Count

Tool File	# of @tool Functions	Tool Names
tools/billing.py	3	get_bill_details , get_payment_history , get_outstanding_balance
tools/plans.py	3	list_available_plans , get_plan_details , com- pare_plans
tools/usage.py	2	get_usage_summary , get_daily_usage
tools/account.py	4	get_customer_profile , lookup_customer_by_name , create_support_ticket , get_open_tickets
tools/tech_support.py	1	search_knowledge_base (RAG-powered)
Total	13	—

Additional supporting module: `tools/rag.py` (RAG vector store builder, not a `@tool` but essential infrastructure).

Tools Verdict:  13 tools across 5 files — well above the 3–5 minimum requirement.

5. Workflow Diagram

Item	Status	Details
workflow_diagram.md	 Present	3,363 bytes — Contains 4 Mermaid diagrams: Architecture Overview, Tool Mapping, Data Flow (sequence diagram), Memory Architecture
README inline diagram	 Present	Mermaid flowchart embedded directly in README.md under the LangGraph Workflow section

Workflow Diagram Verdict:  Standalone diagram file plus inline README diagram both present.

6. Requirements File

Item	Status	Details
requirements.txt	✓ Present	Lists 9 packages across 5 categories: langchain (4), openai, chromadb, gradio, python-dotenv, langsmith

Packages listed:

- langchain>=0.3.0 — Core framework
- langchain-core>=0.3.0 — Core abstractions
- langchain-openai>=0.2.0 — OpenAI integration
- langchain-community>=0.3.0 — Community integrations
- langgraph>=0.2.0 — Graph-based orchestration
- openai>=1.30.0 — OpenAI API client
- chromadb>=0.5.0 — Vector store for RAG
- gradio>=4.40.0 — Web UI framework
- python-dotenv>=1.0.0 — Environment variable management
- langsmith>=0.1.0 — Tracing/observability (optional)

Requirements Verdict: ✓ All necessary dependencies listed with version constraints.

7. Additional Deliverables (Bonus Features)

Feature	Status	File(s)
RAG Integration	✓ Implemented	<code>tools/rag.py</code> , <code>tools/tech_support.py</code> , <code>knowledge_base/*.md</code>
Database Integration (SQLite)	✓ Implemented	<code>data/database.py</code> — 5 tables with seed data
LangSmith Tracing	✓ Implemented	<code>config/settings.py</code> — toggle via <code>LANG-CHAIN_TRACING_V2</code> env var
Streaming Support	✓ Implemented	<code>agents/graph.py</code> — <code>ChatOpenAI(streaming=True)</code>
Conditional Routing	✓ Implemented	<code>agents/graph.py</code> — <code>route_by_intent()</code> with 6 branches
Conversation Memory	✓ Implemented	<code>agents/graph.py</code> — <code>MemorySaver</code> checkpointer
<code>.env.example</code>	✓ Present	Template for required environment variables
<code>.gitignore</code>	✓ Present	Ignores <code>__pycache__</code> , <code>.env</code> , <code>data/telecom.db</code> , <code>data/chroma_db/</code>
Git Repository	✓ Initialized	Initial commit with all files tracked

Summary

Category	Status	Count/Notes
README Required Sections	✓ Complete	10/10 sections present
Screenshots	✓ Complete	7 screenshots in <code>screen-shots/</code>
Demo Video	✓ Present	<code>demo_video.mp4</code> (634 KB)
Tools	✓ Exceeds minimum	13 tools (minimum was 3-5)
Workflow Diagram	✓ Present	<code>workflow_diagram.md</code> + in-line in README
Requirements File	✓ Present	10 packages listed
Bonus Features	✓ Multiple	RAG, SQLite, LangSmith, streaming, conditional routing

Overall Verdict: ✓ All assignment requirements are met.